Part II: Ways to Load Image Datasets

Implement the following six methods for loading the image dataset.

Method 1: TensorFlow Dataset Loader

Load the dataset using **`tf.keras.utils.image_dataset_from_directory()`**.

Required Libraries

```
import tensorflow as tf
import matplotlib.pyplot as plt
```

Tasks

- Load the dataset using `image_dataset_from_directory()`.
- Resize all images to a common size.
- Display a batch of images.
- Print the batch size, image shape, and class names.

Method 2: Keras Image Loader

Load images using **`load_img()`** and convert them into NumPy arrays.

Required Libraries

```
import os
import numpy as np
import matplotlib.pyplot as plt

from tensorflow.keras.utils import load_img, img_to_array
```

Tasks

- Read images one by one.
- Convert images into NumPy arrays.
- Store images and labels separately.
- Display sample images.
- Print the dataset shape.

Method 3: OpenCV

Load the dataset using OpenCV.

Required Libraries

```
import os
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

Tasks

- Read images using `cv2.imread()`.
- Resize all images.
- Convert BGR images to RGB.
- Store images and labels as NumPy arrays.
- Display sample images.

Method 4: Pillow (PIL)

Load the dataset using Pillow.

Required Libraries

```
import os
import numpy as np
import matplotlib.pyplot as plt

from PIL import Image
```

Tasks

- Read images using Pillow.
- Convert images to RGB format.
- Resize images.
- Convert images to NumPy arrays.
- Store the dataset and labels.
- Display sample images.

Method 5: PyTorch Dataset Loader

Load the dataset using **ImageFolder** and **DataLoader**.

Required Libraries

```
import torch
import matplotlib.pyplot as plt

from torchvision import datasets, transforms
from torch.utils.data import DataLoader
```

Tasks

- Create image transformations.
- Load the dataset using `ImageFolder`.
- Create a `DataLoader`.
- Display a sample batch of images.
- Print the batch size and dataset statistics.

Method 6: Custom TensorFlow Data Generator

Create a custom image data generator using `tf.keras.utils.Sequence`.

Required Libraries

```
import os
import numpy as np
import tensorflow as tf
import matplotlib.pyplot as plt

from tensorflow.keras.utils import Sequence, load_img, img_to_array
```

Tasks

- Create a custom data generator class by inheriting from `Sequence`.
- Implement image loading, preprocessing, batching, and label generation.
- Generate batches of images.
- Display sample images from a generated batch.
- Print the batch dimensions.